

CSC11006 - INTRODUCTION TO CLOUD COMPUTING SERVICES

PRACTICE

LAB: Predicting with BigQuery ML

I. General information

ID: LAB
Period: 45 minutes
Deadline:

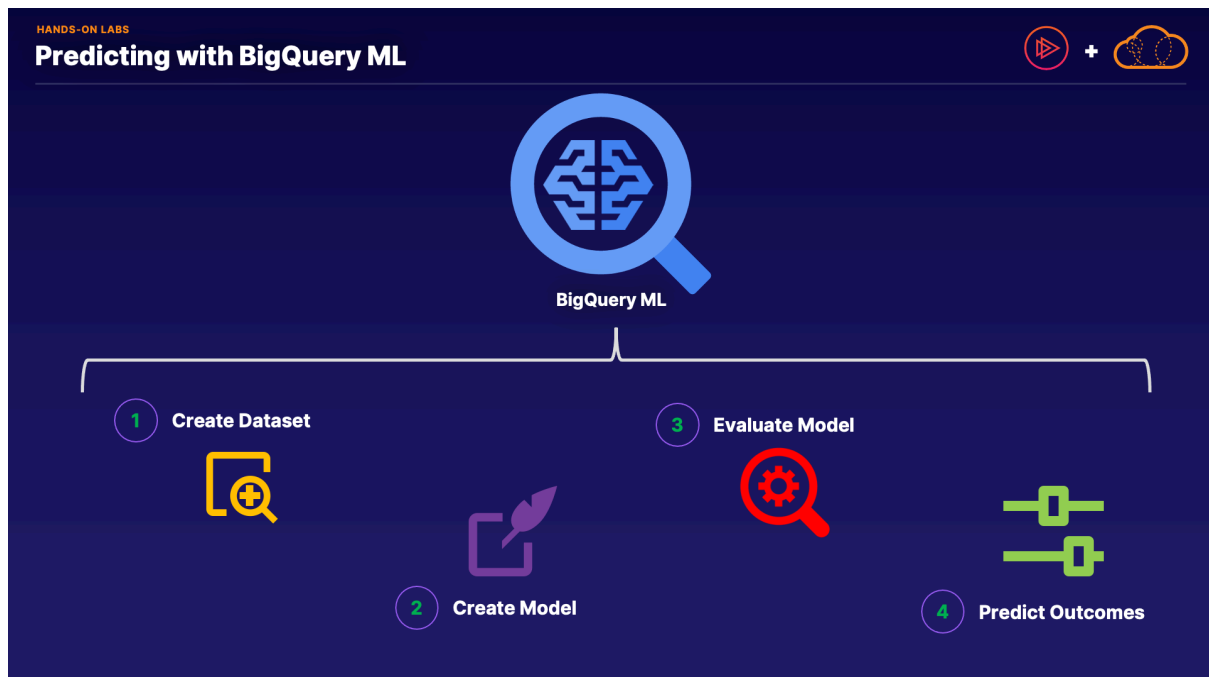
II. Outcome:

This lab will adapt to these following course's outcomes:

- G2.1
- G5.1

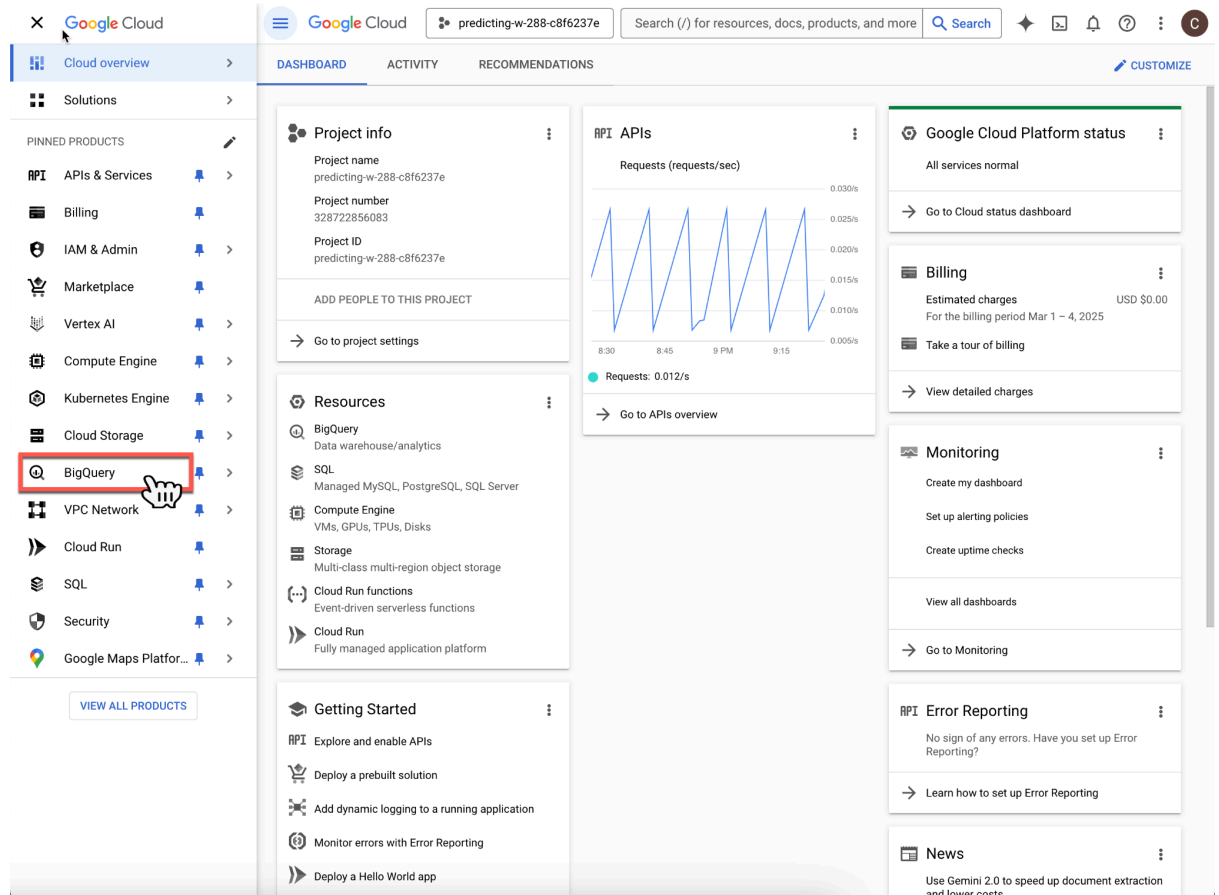
III. Describe:

Machine learning is an incredibly powerful tool that should be accessible to as many users as possible. BigQuery ML brings the power of machine learning to SQL users. In this hands-on lab, you'll leverage BigQuery ML to create, evaluate, and predict the probability that a site visitor will complete a transaction, using a public dataset.

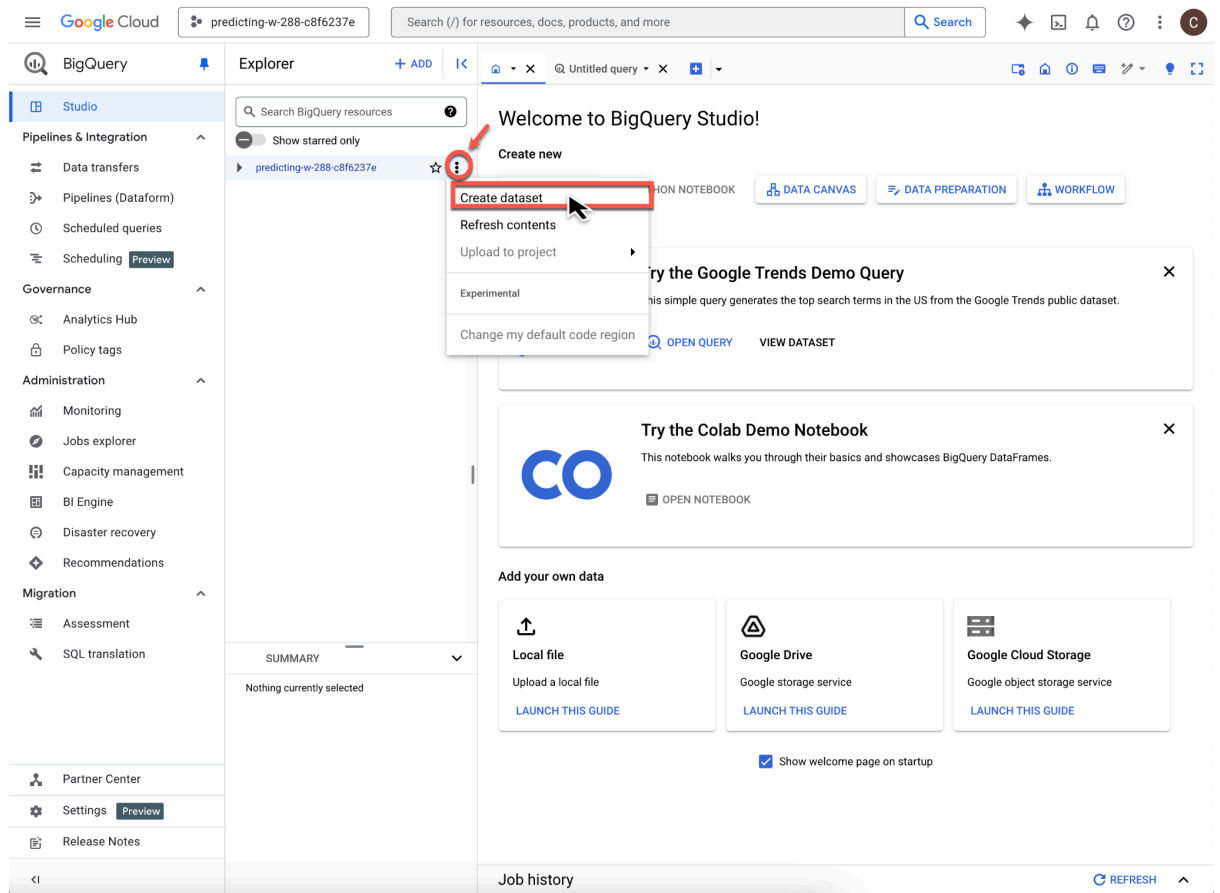


1. Create the BigQuery Dataset

- From the Google Cloud console main navigation (hamburger menu on the top left), select BigQuery. (Click DONE to close the "Welcome" dialog box if it is displayed.)



- Click the vertical three-dot menu next to the pinned project (in the Explorer) to View actions.
- Select Create dataset.



- For Dataset ID, enter acg_bqml.
- For Data location, select us.
- Click CREATE DATASET.

Create dataset

Project ID *

predicting-w-288-c8f6237e

CHANGE

Dataset ID *

acg_bqml

Letters, numbers, and underscores allowed

Location type ?

☐ Region

Specify a region to colocate your datasets with other Google Cloud services.

☒ Multi-region

Allow BigQuery to select a region within a group to achieve higher quota limits.

Some locations have been restricted due to a policy set by your organization. [Learn more about restricting locations.](#)

Multi-region *

US (multiple regions in United States)

External Dataset

The selected region supports the following external dataset types: Cloud Spanner

☐ Link to an external dataset ?

Tags

Advanced options

CREATE DATASET

CANCEL

2. Create and Train the ML Model

- Add the following query to the Query editor. This query first creates and trains the model with the dataset/table `acg_bqml.transactions_model`. The query then declares the `model_type` as a logistic regression. The query selects the columns declared within the `SELECT` statement. Data is selected from the public `bigquery-public-data.google_analytics_sample.ga_sessions_*` dataset, where the `*` indicates you'll query a set of tables sharded by date. The `WHERE` clause indicates that tables between August 1, 2016 and June 30, 2017 will be queried:

```
CREATE MODEL `acg_bqml.transactions_model`  
OPTIONS(model_type='logistic_reg') AS
```

```

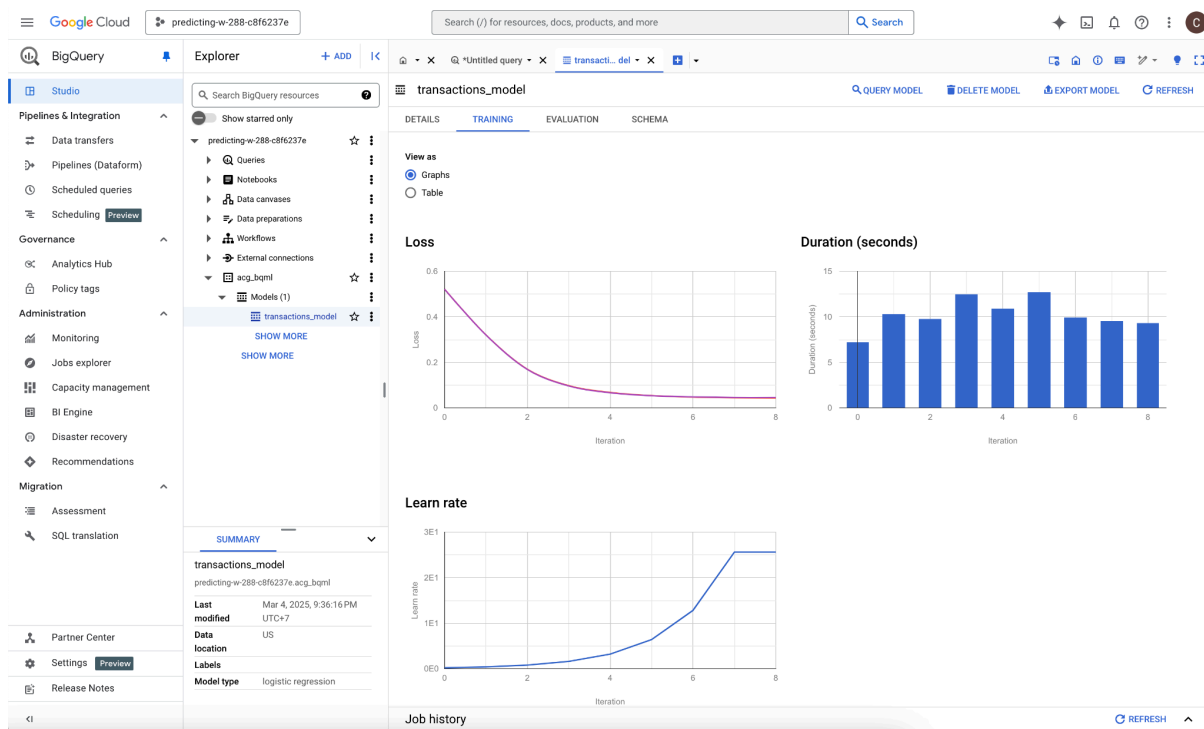
SELECT
  IF(totals.transactions IS NULL, 0, 1) AS label,
  IFNULL(device.operatingSystem, "") AS os,
  device.isMobile AS is_mobile,
  IFNULL(geoNetwork.country, "") AS country,
  IFNULL(totals.pageviews, 0) AS pageviews
FROM
  `bigquery-public-data.google_analytics_sample.ga_sessions_*`
WHERE
  _TABLE_SUFFIX BETWEEN '20160801' AND '20170630'

```

- At the top, click RUN. It will take a minute or two for the query to process.

The screenshot displays the Google Cloud BigQuery Studio interface. On the left is a navigation sidebar with categories like Pipelines & Integration, Governance, Administration, and Migration. The main area is split into two panes. The top pane, titled 'Untitled query', contains the SQL query from the previous block. The bottom pane, titled 'Query results', shows the 'EXECUTION GRAPH' tab. This graph illustrates the execution flow of the query, with three stages: 'Validate' (duration 26 sec), 'Preprocess' (duration 25 sec), and 'Train'. Each stage is represented by a colored box with a green checkmark indicating successful completion. The 'Train' stage is highlighted with a blue border. Above the graph, there are tabs for 'JOB INFORMATION', 'RESULTS', 'EXECUTION DETAILS', and 'EXECUTION GRAPH'. Below the graph is a 'Job history' section with a 'REFRESH' button.

- In the Results tab at the bottom, click GO TO MODEL.
- Select the TRAINING tab. Observing the Loss is decreasing, while the Learn rate is increasing.



3. Evaluate the ML Model

- Go back to the Query editor, and replace the existing query with the query below. This query will evaluate the model and is similar to the previous query.
Note that the SELECT statement will include all fields from the evaluated model. Additionally, observe the inclusion of the ML.EVALUATE statement, which will evaluate the newly created model:

SELECT

*

FROM

ML.EVALUATE(MODEL `acg\_bqml.transactions\_model`, (

SELECT

IF(totals.transactions IS NULL, 0, 1) AS label,

IFNULL(device.operatingSystem, "") AS os,

device.isMobile AS is_mobile,

IFNULL(geoNetwork.country, "") AS country,

IFNULL(totals.pageviews, 0) AS pageviews

FROM

`bigquery-public-data.google_analytics_sample.ga_sessions_*

WHERE

_TABLE_SUFFIX BETWEEN '20170701' AND '20170801'))

- At the top, click RUN.
- In the Results tab at the bottom, observe the fields returned, which evaluate the model's accuracy and precision.

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Search (/) for resources, docs, products, and more

BigQuery Explorer

Search BigQuery resources

predicting-w-288-c8f6237e

Queries

Untitled query

```

1 SELECT
2 +
3 FROM
4 ML_EVALUATE(MODEL `acg_bqml.transactions_model`, (
5 SELECT
6 IF(totals.transactions IS NULL, 0, 1) AS label,
7 IFNULL(device.operatingSystem, "") AS os,
8 device.isMobile AS is_mobile,
9 IFNULL(geoNetwork.country, "") AS country,
10 IFNULL(totals.pageviews, 0) AS pageviews
11 FROM
12 `bigquery-public-data.google_analytics_sample.ga_sessions_*`
13 WHERE
14 _TABLE_SUFFIX BETWEEN '20170701' AND '20170801'))

```

Query results

JOB INFORMATION RESULTS CHART JSON EXECUTION DETAILS EXECUTION GRAPH

Row	precision	recall	accuracy	f1_score	log_loss	roc_auc
1	0.468503937007...	0.110800744878...	0.985343158347...	0.179216867469...	0.046242211011...	0.981746253746...

Results per page: 50 1 - 1 of 1

Job history

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Search (/) for resources, docs, products, and more

BigQuery Explorer

Search BigQuery resources

predicting-w-288-c8f6237e

Models (1)

transactions_model

DETAILS TRAINING EVALUATION SCHEMA

Aggregate Metrics

Threshold	0.5000
Precision	0.3860
Recall	0.1803
Accuracy	0.9867
F1 score	0.2458
Log loss	0.0454
ROC AUC	0.9788

Score threshold

Positive class threshold	0.0015
Positive class	1
Negative class	0
Precision	0.0121
Recall	1.0000
Accuracy	0.0121
F1 score	0.0238

Precision-recall by threshold

Precision-recall curve

ROC curve

Confusion matrix

Job history

4. Predict Using the ML Model

- Back up in the Query editor, replace the existing query with the query below to make predictions. The initial SELECT statement retrieves the country column and computes a sum of the predicted_label column from the ML.PREDICT function. The nested SELECT, FROM, and WHERE clauses are repeated, with the results grouped by country and ordered by total_predicted_purchases:

```

SELECT
  country,
  SUM(predicted_label) as total_predicted_purchases
FROM
  ML.PREDICT(MODEL `acg_bqml.transactions_model`, (
SELECT
  IFNULL(device.operatingSystem, "") AS os,
  device.isMobile AS is_mobile,
  IFNULL(totals.pageviews, 0) AS pageviews,
  IFNULL(geoNetwork.country, "") AS country
FROM
  `bigquery-public-data.google_analytics_sample.ga_sessions_*`
WHERE
  _TABLE_SUFFIX BETWEEN '20170701' AND '20170801'))
GROUP BY country
ORDER BY total_predicted_purchases DESC
LIMIT 15

```

- At the top, click RUN.
- In the Results tab at the bottom, observe the total_predicted_purchases column in the output. Note the United States has the most predicted purchases, followed by Taiwan, and then Canada.

The screenshot displays the Google Cloud BigQuery Studio interface. The top navigation bar includes the Google Cloud logo, a project ID 'predicting-w-288-c8f6237e', and a search bar. The left sidebar contains navigation links for Studio, Pipelines & Integration, Governance, Administration, and Migration. The main workspace is divided into three panes: Explorer, Query Editor, and Query Results.

The Explorer pane shows a project structure with folders for Queries, Notebooks, Data canvases, Data preparations, Workflows, External connections, and Models. The 'Models' folder is expanded, showing a model named 'transactions_model'.

The Query Editor pane displays the following SQL query:

```

1 SELECT
2   country,
3   SUM(predicted_label) as total_predicted_purchases
4 FROM
5   ML.PREDICT(MODEL `acg_bqml.transactions_model`, (
6 SELECT
7   IFNULL(device.operatingSystem, "") AS os,
8   device.isMobile AS is_mobile,
9   IFNULL(totals.pageviews, 0) AS pageviews,
10  IFNULL(geoNetwork.country, "") AS country
11 FROM
12   `bigquery-public-data.google_analytics_sample.ga_sessions_*`
13 WHERE
14   _TABLE_SUFFIX BETWEEN '20170701' AND '20170801'))
15 GROUP BY country
16 ORDER BY total_predicted_purchases DESC
17 LIMIT 15

```

The Query Results pane shows the output of the query. The results are displayed in a table with columns 'country' and 'total_predicted_purchases'. The table is sorted by 'total_predicted_purchases' in descending order. The first three rows are highlighted with a red box:

Row	country	total_predicted_purchases
1	United States	220
2	Taiwan	8
3	Canada	7
4	India	2
5	Turkey	2
6	Japan	2
7	Indonesia	1
8	Serbia	1
9	Venezuela	1
10	Singapore	1
11	Germany	1
12	Thailand	1
13	Brazil	1

The bottom of the interface shows the 'Job history' section with a 'REFRESH' button.